

PROJECT ON STRAVA'S CUSTOMER ANALYSIS

Final Dataset Overview: The Master Fact Table

1. Dataset Purpose & Architecture

The final dataset (Fitbit_Business_Master_Hourly) serves as the central "Source of Truth" for the entire analytics project. It is a highly optimized, wide-format Fact Table resulting from the strategic consolidation of **8 distinct raw datasets**, processed via Python and pushed directly into a Microsoft SQL Server database (PRACTICEDB) for dynamic Tableau dashboarding.

Crucially, the data is structured at the **Hourly Grain** (one unique row per user, per hour). To achieve this without losing the value of highly granular intraday datasets (like second-by-second heart rate or minute-by-minute METs), micro-metrics were mathematically aggregated and rolled up. Simultaneously, broad daily metrics (like total sleep or weight) were seamlessly mapped onto this hourly timeline without causing duplicate Cartesian explosions.

2. Key Data Categories (Feature Dictionary)

The dataset consists of 28 distinct columns, logically grouped into four core business categories:

A. Identification & Timeline (The Primary Keys) * Id: The unique 10-digit identifier for each user.

- **Datetime_Hour:** The specific hour of the day the tracked events occurred (Native SQL Datetime format).

B. Intraday Exertion & Biometrics (The Hourly Grain) This category tracks the specific physical output of the user on an hour-by-hour basis.

- **Hourly Activity:** HourlyCalories, HourlySteps, HourlyTotalIntensity, HourlyAvgIntensity.
- **Aggregated Biometrics:** Avg_HR (Rolled up from second-level data), Avg_METs (Rolled up from minute-level data).

C. Daily Activity Context (Left-Joined) This maps the user's overall daily performance to give context to their hourly efforts.

- **Movement & Distance:** TotalSteps, TotalDistance, VeryActiveDistance, ModeratelyActiveDistance, LightActiveDistance, SedentaryActiveDistance.
- **Time by Intensity:** VeryActiveMinutes, FairlyActiveMinutes, LightlyActiveMinutes, SedentaryMinutes.
- **Energy Output:** DailyCalories.

D. Recovery & Body Composition (Left-Joined) Tracks static health metrics and rest periods.

- **Sleep Metrics:** TotalSleepRecords, TotalMinutesAsleep, TotalTimeInBed.
- **Body Metrics:** WeightKg, WeightPounds, BMI.
- **Behavioral Flag:** IsManualReport (Indicates if the user manually logged weight vs. auto-syncing).

3. Analytical Value & Business Application

By flattening 8 disjointed tables into this single, hourly Master Fact Table, the architecture is perfectly optimized for Tableau's processing engine.

This structure allows analysts to track **intraday trends** (e.g., "At what hour do high-BMI users burn the most calories?") while simultaneously running correlations against daily totals (e.g., "How does a poor night of TotalMinutesAsleep impact the next day's HourlyTotalIntensity at 8:00 AM?").

Project Methodology: Data Architecture and ETL Pipeline

How the Data Was Engineered in Python

- **Aggregating the Granular:** Second-level heart rate and minute-level METs were grouped and averaged up to the "Hourly" level using Pandas datetime flooring (`dt.floor('H')`) to create a consistent timeline.
- **Preventing Column Collisions:** Duplicate column names across files (e.g., changing 'Calories' to 'HourlyCalories' and 'DailyCalories') were renamed prior to joining to maintain a clean schema.
- **The Hourly Outer Join:** Merged all hourly datasets together using a FULL OUTER JOIN based on Id and Datetime_Hour. This ensured no data was dropped if a user tracked heart rate but not steps during a specific hour.
- **Pre-Deduplication (Preventing Data Bloat):** Applied a strict `drop_duplicates()` logic to the daily context files *before* merging. This prevented a many-to-many Cartesian explosion if a user synced their app twice in one day.
- **The Daily Left Join:** Extracted the standard Date from the hourly timestamps, mapping the daily context metrics (Sleep, Weight, Total Activity) to their corresponding hours via LEFT JOIN.
- **Smart Imputation:** Handled missing data logically based on context. Activity gaps were zero-filled (0). Missing heart rates utilized a cascading failsafe: filling with the user's daily average, then all-time average, and finally a standard 70 BPM resting floor. Weight and sleep metrics utilized forward/backward filling (`ffill/bfill`).

Dataset Selection Strategy

To ensure a highly performant and normalized database, **10 out of the 18 original files were intentionally discarded** during the ETL process.

Accepted Datasets (8 Files Used in the Pipeline)

The Hourly Base (3 Datasets): * hourlyCalories_merged.csv

- hourlySteps_merged.csv
- hourlyIntensities_merged.csv

The Granular Roll-ups (2 Datasets): * heartrate_seconds_merged.csv

- minuteMETsNarrow_merged.csv

The Daily Context (3 Datasets): * dailyActivity_merged.csv

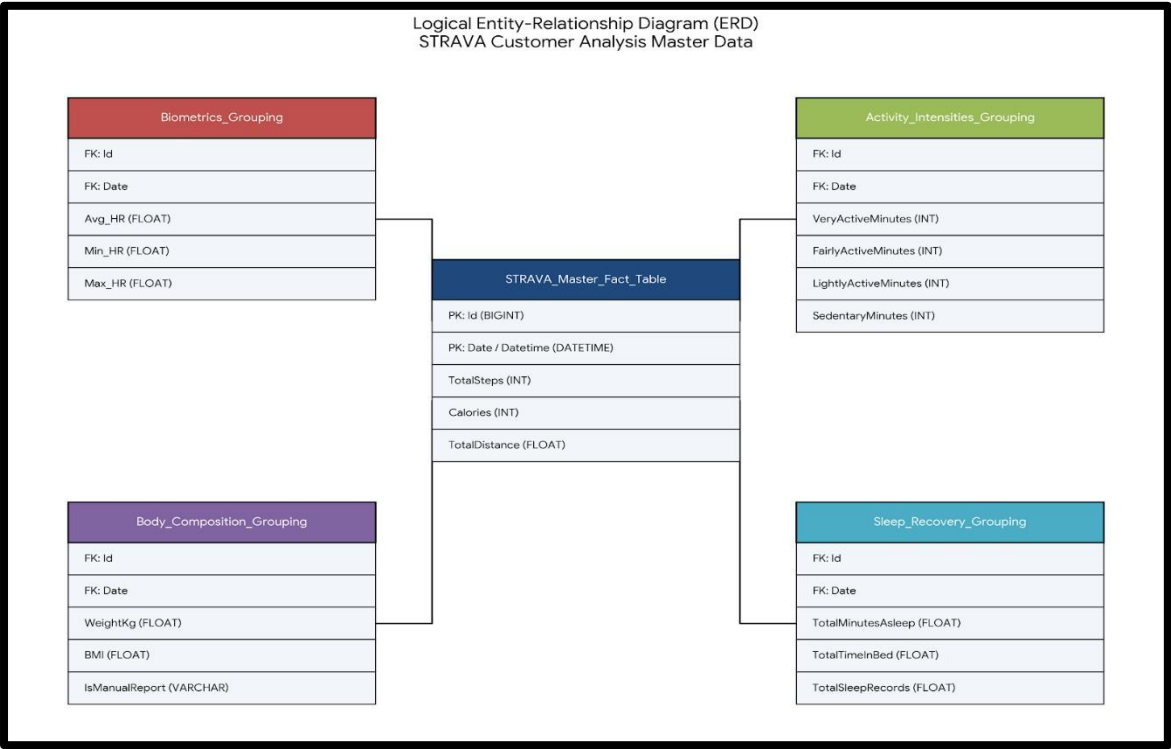
- sleepDay_merged.csv
- weightLogInfo_merged.csv

Discarded Datasets (10 Files Excluded)

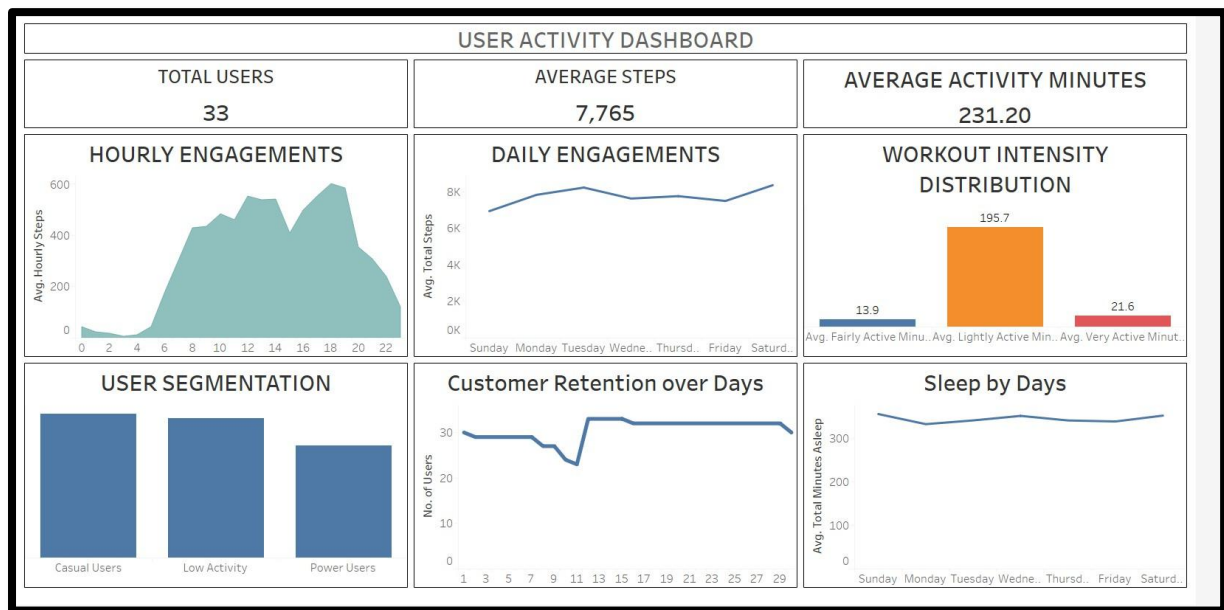
These files were strategically excluded to prevent redundant data bloat and messy joins:

- **The Redundant Daily Files (3):** dailyCalories, dailyIntensities, dailySteps.
 - *Reason:* Pre-aggregated redundancy. All exact metrics in these files were already consolidated inside the master dailyActivity_merged.csv.
- **The Redundant Minute Files (6):** minuteCalories, minuteSteps, minuteIntensities (Both Wide & Narrow formats).
 - *Reason:* Granularity duplication & Schema violations. The "Wide" formats violate database normalization rules. Furthermore, the native "Hourly" files already contained the exact sum of these metrics. Re-aggregating the narrow minute files would waste compute power.
- **The Noisy Sleep File (1):** minuteSleep_merged.csv.
 - *Reason:* Too granular. Minute-by-minute sleep states (awake, restless, asleep) introduce unnecessary complexity for a high-level KPI dashboard. The sleepDay_merged.csv provided the exact summary metrics needed for business analysis.

ENTITY RELATIONSHIP DIAGRAM(ERD):



BUSINESS ANALYSIS:



1. Top-Line KPIs: The Capacity for Growth

- **The Insight:** The tracked cohort (n=33) averages 7,765 steps and 231 active minutes daily, placing them 22% below the industry-standard 10,000-step health benchmark.
- **The "So-What?" (Business Impact):** We are currently operating as a "Passive Tracking" utility for casual movers rather than an "Active Improvement" tool.
- **Strategic Action:** Our primary product goal must be to bridge the "Activity Gap." We need to introduce achievable gamification that rewards the conversion of sedentary time into active steps, thereby increasing daily app reliance and overall Lifetime Value (LTV).

2. Time-Based Patterns: Optimizing Marketing ROI

- **The Insight:** User activity follows a distinct dual-peak rhythm (11:30 AM – 12:30 PM and 16:00 – 20:00) with overall engagement volume peaking on Saturdays.
- **The "So-What?" (Business Impact):** Users are clearly constrained by sedentary weekday work schedules and only engage during lunch breaks, evening commutes, and weekends.
- **Strategic Action:** Establish strict "High-Priority Notification Windows." Marketing must synchronize daily summaries and push notifications precisely at 11:30 AM and 16:30 PM. Furthermore, we should launch "Weekend Social Challenges" to leverage the Saturday momentum for organic, viral community growth.

3. Quality of Engagement: The Intensity Shift

- **The Insight:** When filtering out sedentary noise, the vast majority of active time is logged as "Lightly Active," offering a very low metabolic ROI for the user.

- **The "So-What?" (Business Impact):** The app's current value proposition is skewed entirely toward passive observation. If we can shift users toward "Very Active" bursts, we massively increase the product's perceived health-utility value, which drives premium subscription conversions.
- **Strategic Action:** Deploy UI nudges that promote 10-minute "Micro-Burst" challenges (e.g., a brisk post-lunch walk) to systematically convert light activity into moderate/vigorous health milestones.

4. Domain Context & External Drivers (The Day 12 Anomaly)

- **The Insight:** Daily Active User (DAU) retention shows a severe dip on Day 11, immediately followed by a massive platform surge on Day 12 (May 12th).
- **The "So-What?" (Business Impact):** May 12th is *International Nurses Day*. This anomaly strongly suggests our core "Power User" segment is heavily comprised of healthcare professionals. The Day 11 dip reflects pre-shift preparation, while Day 12 is a reaction to a domain-specific milestone.
- **Strategic Action:** Validate this hypothesis by cross-referencing user profession data. If confirmed, unlock a targeted B2B marketing track or sponsorship campaign focusing specifically on healthcare workers to dominate this high-performing niche.

5. Biological Churn: The Sleep Debt Correlation

- **The Insight:** The cohort averages only ~350 minutes (5.8 hours) of sleep per night—dangerously below the 7–8 hour health benchmark.
- **The "So-What?" (Business Impact):** There is a direct, undeniable link between this "Sleep Debt" and the high volume of "Casual" users. Users aren't lazy; they are physiologically under-rested, which limits their capacity for high-intensity workouts and drives mid-week app churn.
- **Strategic Action:** Build a "Recovery Module." Show users exactly how an extra 30 minutes of sleep increases their step count the following day. This pivots the app from just a fitness tracker into a holistic health partner.

6. Behavioural Evolution: Segmentation Strategy

- **The Insight:** "Casual" and "Low Activity" users form the bulk of the population, while a small fraction logs upwards of 16,000 steps.
- **The "So-What?" (Business Impact):** Our immediate revenue lever is *Segment Migration*.
- **Strategic Action:** *For Low Activity:* Implement "Micro-nudges" and streak-protectors to prevent total churn.
 - *For Power Users:* Launch a "VIP Ambassador" program. Leverage these 16k-step users as the faces of our community social challenges to inspire and pull the "Casual" segment into higher engagement tiers.